

# Lineside Fires

Module 3 | Rule Book: G1 §1.6 | Source: PPT 25-03

## 1. High Fire Risk Zone Signs

Areas of woodland, heather or dry scrubland are marked with circular yellow lineside signs:

| Start of Zone                             | End of Zone                               |
|-------------------------------------------|-------------------------------------------|
| Yellow circle — black conifer tree symbol | Yellow circle — single vertical black bar |

## 2. Action if You See a Lineside Fire

### Rule: G1 §1.6

Immediately report the lineside fire to the signaller or Operations Control.

If OLE could be affected — also contact the ECO to arrange isolation of the electrical supply (AC and DC rule books).

For a fire not involving your train — call the fire brigade (999). Never assume someone else has already called.

## 3. Forms of Ignition — Railway Examples

PPT 25-03 (Slide 4) identifies Direct, Electrical and Friction as the most likely causes. Handout 25a covers all seven types. Key four:

| Type                   | Railway Example                                                                                                                              |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Direct Ignition</b> | Lit cigarette dropped on a fabric seat; sparks from trackside welding landing on dry vegetation                                              |
| <b>Friction</b>        | Seized wheel bearing overheating; wheel flats producing sustained sparks; rail grinding trains                                               |
| <b>Radiation</b>       | Lineside grass fire heating wooden sleepers across an air gap; burning vehicle radiating heat through boundary fencing                       |
| <b>Conduction</b>      | Electrical fault conducting heat through floor panel into insulation; heat from burning bogie conducting along axle into adjacent components |

## 4. Three Ways Fire Spreads

PPT 25-03 Slide 14. Fire doors must be used correctly to limit spread.

| Type              | Railway Example                                                                                                                                         |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Conduction</b> | Heat travels through solid materials — burning seat frame conducts heat along metal brackets into floor, igniting carpet and insulation beyond the seat |

|                   |                                                                                                                                                                                                               |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Convection</b> | Hot gases and smoke rise and bank down — the biggest spreader in an enclosed carriage; smoke fills from the ceiling down, heating everything above head height and igniting overhead racks and ceiling panels |
| <b>Radiation</b>  | Heat energy travels across an air gap — a burning seat can ignite the seat directly opposite across the aisle without the flames ever making contact                                                          |

## 5. Impact on Railway Infrastructure & Environment

| Area                             | What can be affected                                                                                                                                                                              |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Track</b>                     | Wooden sleepers burn — loss of support, misalignment. Rails expand/buckle from heat. Ballast cracks and loses drainage properties.                                                                |
| <b>Earthworks</b>                | Vegetation burns off embankments, destabilising the structure. Risk of slippage or erosion in rain.                                                                                               |
| <b>Signalling &amp; Telecoms</b> | Lineside cables burn in cable troughs — signal failures, loss of track circuits. Signal posts and equipment cabinets destroyed. GSM-R links severed.                                              |
| <b>OLE</b>                       | Wooden OLE masts can catch fire causing mast collapse and wire drop. Requires ECO to isolate supply before recovery. Smoke and heat affect wire tension.                                          |
| <b>Structures</b>                | Wooden station canopies, platform furniture, fencing, older bridge timbers.                                                                                                                       |
| <b>Environment</b>               | Destruction of lineside habitats (bats, nesting birds). Fire retardant chemicals contaminate watercourses. Ash blocks drainage. Fire can spread beyond railway boundary into farmland or housing. |
| <b>Operations</b>                | Line closed for inspection — extended if OLE involved. Speed restrictions until track geometry confirmed safe. Bus replacement required.                                                          |

*Rule Book References: G1 §1.6 (Reporting lineside fires) | M1 §4 (Fire on a train) | AC §6.1.1 / DC §6.1.1 (OLE isolation)*